








Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



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Why Do Assets Have Different Average Returns?

What we will explain

- What explains expected returns? The Capital Asset Pricing Model (CAPM) and multifactor models
- How are exposures estimated? Betas, their uncertainty and shrinkage (pulling a noisy estimate towards a target value)
- How are pricing restrictions tested? A joint test of the alphas, Fama–MacBeth regressions, model comparison
- Do the findings survive? Multiple testing, latent factors, replication

What you should be able to do

- estimate a beta with heteroskedasticity- and autocorrelation-robust errors and shrink it
- test the CAPM alphas jointly and estimate risk premia with robust errors
- judge a new factor against multiple-testing hurdles and replication evidence

Route through the Chapter

1. What Explains Expected Returns?
2. From Markowitz to the CAPM
3. Estimating Beta, with a quick check
4. Testing the CAPM, with a quick check
5. Multifactor Models
6. Fama–MacBeth Regressions, with a quick check
7. Discount Factors and Model Comparison
8. The Factor Zoo and Multiple Testing, with a quick check
9. Latent Factors: PCA and IPCA
10. Factor Investing
11. Case Study: Is There a Replication Crisis?
12. AI (Artificial Intelligence) for Scientific Discovery
13. Conclusions

- **First pass**: Sections 1–6, 8 and 10, the results of Section 11 and the conclusions; answer each quick check before opening its answer in the Appendix
- **Deeper study**: slides marked *(deeper)* in the title (the generalized method of moments, useless factors, discount factors, number of factors, instrumented PCA, the Bayesian hierarchy of the case study) and the Appendix, reached through the buttons on the slides
- **Practice**: each method has a Seminar 3 problem, named and linked on the slide that uses it
- Section 12 prepares the course project

What Is a Factor Model?

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \ldots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1} f_{1,t} + \cdots + \beta_{i,K} f_{K,t} + \varepsilon_{i,t}$$

- $R_{i,t}$: return of asset i in period t
- $R_{f,t}$: risk-free rate
- K: number of factors
- α_i: average return not explained by the factors, 0 if the factors price the asset correctly
- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k
- $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock

What Is a Factor Model Used For?

- Two uses in this chapter
  - Explain an asset: the technology ETF (exchange-traded fund) XLK moves 1.28 times the US market (Section 3)
  - Evaluate a fund: is its α_i positive once its exposures are priced (Sections 4 and 10)?
- Other uses: cost of capital, portfolio construction and risk management (Campbell, Lo & MacKinlay (1997), chapters 5–6)
- The risk-management advantage: N assets have $N(N+1)/2$ variances and covariances, while a K-factor model needs only the loadings, the factor covariance and N specific variances
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

The Data: US Factors, ETFs and the Bucharest Stock Exchange

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size × B/M portfolios	US stocks, two sorts	1963–2026
US sector ETFs	S&P 500 sectors: 9 from 1999, 11 from 2018	1999–2026
Factor ETFs	momentum, value, size, quality, minimum volatility	2013–2026
BVB blue chips, BET	11 stocks and their index	2015–2026

- Sector ETF tickers: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP consumer staples, XLU utilities, XLV health care, XLY consumer discretionary
- From 2018 also XLRE real estate and XLC communication services

- Factors and portfolios: Kenneth French Data Library, monthly returns in percent, to July 2026 (momentum: to August 2026)
- ETFs and stocks: daily adjusted closing prices (dividends reinvested), to 18 September 2026
- **S&P 500**: the index of 500 large US companies
- **BVB**: Bursa de Valori București, the Bucharest Stock Exchange; **BET**: its benchmark index, a price index (no dividends)
- **B/M**: book value of equity over its market value
- **Excess return** $R^e_{i,t} = R_{i,t} - R_{f,t}$; $R_{f,t}$: one-month Treasury bill rate (RF in the French data)
- **Mkt-RF**: excess return of the US stock market, weighted by market capitalisation (value-weighted), from the same data

Simple or Log Returns?

- **Simple return** (the percentage price change): $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$
  - $P_{i,t}$: adjusted price (dividends reinvested) of asset i at the end of period t
- **Log return**: $r_{i,t} = \ln(P_{i,t}/P_{i,t-1}) = \ln(1 + R_{i,t})$
- Example: the price rises from 100 to 110, so $R = 10\%$, but $r = \ln 1.1 = 9.53\%$
- If the price falls back to 100: $R = -9.09\%$ and $r = -9.53\%$, so the two log returns sum to 0
- Log returns add over time, simple returns add across assets: $R_{p,t} = \sum_i w_i R_{i,t}$ (Chapter 0)
- The French factors are portfolio returns, so regressions on them use simple returns
- Rolling betas, BVB betas and principal component analysis (Section 9) use daily log returns
- BVB data cleaning drops days with $|r_t| > 0.5$: a one-day rise above 64.9% ($e^{0.5} - 1$) or a one-day fall above 39.3% ($1 - e^{-0.5}$)
- At the BVB the daily risk-free rate is treated as a constant: a constant changes no covariance, so beta is unaffected

The Mean–Variance Frontier

- Markowitz (1952): with weights w, expected excess returns $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, the frontier solves

$$\min_{\mathrm{w}} \ \mathrm{w}^{\top} \boldsymbol{\Sigma}\, \mathrm{w} \quad \text{subject to} \quad \mathrm{w}^{\top} \boldsymbol{\mu} = \mu^{*},\ \mathrm{w}^{\top} 1 = 1$$

  - in words: the lowest variance for a target mean μ^{*}
  - 1: a vector of ones, so the weights sum to 1
  - $^{\top}$: transpose
- Negative weights are **short positions** (short sales: the asset is borrowed and sold)

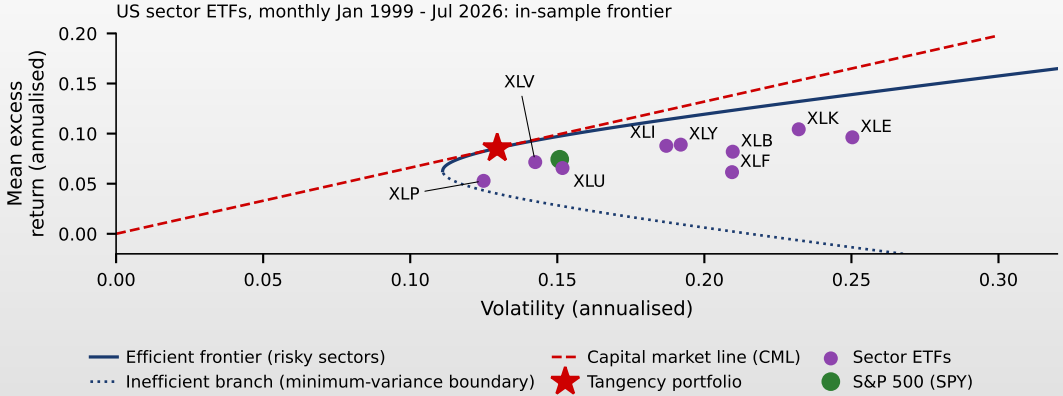
The Sharpe Ratio and the Tangency Portfolio

- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f]/\sigma_p$: the excess return per unit of volatility
- **Tangency portfolio**: the portfolio with the highest Sharpe ratio (the formula below holds when $\boldsymbol{\Sigma}$ is positive definite and the denominator is positive)

$$w_{\text{tan}} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \,/\, 1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$$

  - R_p, σ_p: return and volatility of portfolio p
  - the denominator rescales $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ so that the weights sum to 1
- **CML** (Capital Market Line): all mixes of the risk-free asset and the tangency portfolio, $\mu_p = SR_{\text{tan}}\,\sigma_p$
- From the tangency portfolio to the beta pricing equation: Seminar 3, problem A2
- Estimation error and shrinkage: Chapter 4

The Efficient Frontier of US Sectors (1/2)



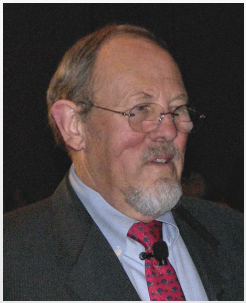
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The Efficient Frontier of US Sectors (2/2)

- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)
- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the ETF that tracks the S&P 500, used as the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49
- The CAPM requires the market to lie on the frontier; in this sample SPY does **not**
- The chart axes are annualised fractions: 0.10 on an axis means 10%
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- The frontier is computed **in-sample** from very uncertain means (Chapter 4)
- Is the gap between SPY and the frontier larger than estimation noise? The GRS test (Gibbons–Ross–Shanken, Section 4) on What a CAPM Test Establishes says no ($p = 0.85$)

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The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics 1990

Photo: Larry D. Moore (2007); CC BY 4.0; Wikimedia Commons

- The **CAPM** was formulated by Sharpe (1964) and Lintner (1965)
- If all investors choose mean–variance efficient portfolios and share the same beliefs, they all hold the same tangency portfolio
- Since every asset must be held by someone, the tangency portfolio is the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i \left(\mathbb{E}[R_m] - R_f\right), \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

- R_i, R_m: returns of asset i and of the market portfolio
- $\mathbb{E}[R_m] - R_f$: the **market risk premium**
- β_i: sensitivity of asset i to the market; $\beta_i > 1$ amplifies market moves
- Black (1972): a version without unlimited borrowing at the risk-free rate, the zero-beta CAPM (Section 4)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

What the CAPM Implies

- Main assumptions
  - Mean–variance investors with identical beliefs, one period
  - Unlimited borrowing and lending at R_f
  - No taxes or trading costs
- Only beta is priced: no other characteristic should predict returns
- Idiosyncratic risk earns nothing
- **SML** (Security Market Line): the testable statement of the CAPM (Section 4)
  - every asset lies on the line (beta, mean), with slope equal to the market premium
  - the CML contains only efficient portfolios
- The CAPM cost of capital is an estimated beta times an estimated market premium; the uncertainty of the premium weighs far more than that of beta
  - XLK example: the standard error (s.e.) of the market premium is $15.4\%/\sqrt{63} = 1.9\%$ a year (15.4%: annual market volatility; 63 years of data); times $\hat{\beta} = 1.28$ this gives 2.5%
  - The uncertainty of beta contributes only $0.082 \times 7.2\% = 0.6\%$ (the s.e. of beta times the premium)
- Which portfolio is "the market"? The answer decides what a test of the CAPM can establish (Roll's critique, Section 4)

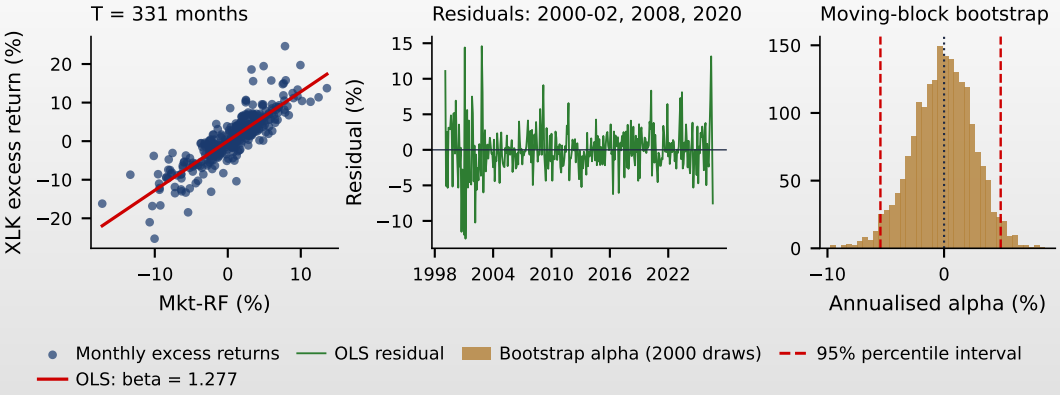
The Market Model and HAC Errors

- Market model $R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$
- **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\mathrm{Cov}}(R^e_i, R^e_m)/\widehat{\mathrm{Var}}(R^e_m)$
- Variance split $\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma^2_{\varepsilon,i}$: systematic risk (priced under the CAPM) plus idiosyncratic risk (diversifiable)
  - σ_i^2, σ_m^2: variances of the asset and of the market; $\sigma^2_{\varepsilon,i}$: residual variance
  - $R^2 = \beta_i^2\sigma_m^2/\sigma_i^2$ is the systematic share
- Volatility clustering (Chapter 1) makes residuals heteroskedastic: use **HAC** (heteroskedasticity and autocorrelation consistent) errors (Newey & West (1987))
  - Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags (Newey & West (1994))
  - T: number of periods, $\lfloor \cdot \rfloor$: integer part
  - $T = 331$ gives $L = 5$; $T = 757$ gives $L = 6$
  - the point estimates remain those of OLS

Choices to Report for Any Beta

- Each of these choices can change the estimated beta
  - Frequency and window: 3–5 years of monthly data or 1–2 years of daily data
  - Daily data suffer from asynchronous trading: the closing prices of different assets are not recorded at the same moment
  - Market proxy: SPY (S&P 500) or Mkt-RF for US assets, BET for Romanian stocks
  - Data cleaning: a scale error creates a fake return
  - Example: for Digi Communications (ticker DIGI), the closes of 17–22 May 2017 are about 725 times those from 23 May on, which creates a fake one-day log return of -6.6

Worked Example: Beta of the Technology ETF (1/2)



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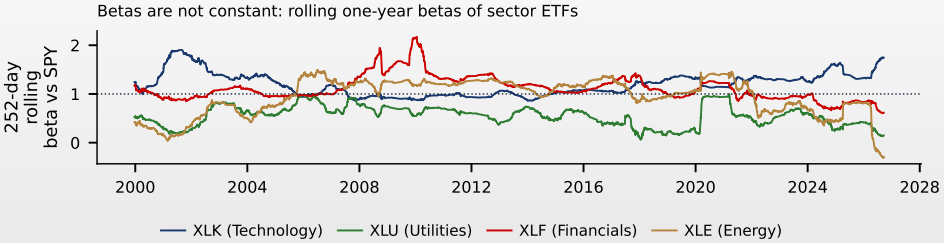
Worked Example: Beta of the Technology ETF (2/2)

		Standard error		
	Estimate	**Classical**	**White**	**Newey–West**
α (annual)	0.33%	2.27%	2.32%	2.45%
β	1.277	0.041	0.054	0.082

- XLK on Mkt-RF, monthly 1999–2026, $T = 331$, $R^2 = 0.742$
- **Classical**: errors assumed homoskedastic and serially uncorrelated
- **White**: robust to heteroskedasticity (White (1980))
- **Newey–West**: also robust to autocorrelation, $L = 5$ lags
- The robust s.e. of β is twice the classical one
- Heteroskedasticity explains part of the increase (White column); autocorrelation of the product $f_t\varepsilon_t$ explains the rest
- t-statistic of alpha: $t(\alpha) = 0.33/2.45 = 0.13$
- 95% bootstrap interval for alpha: $[-5.5\%, 4.9\%]$
  - moving-block bootstrap: blocks of 12 consecutive months of the pairs (R^e_t, f_t) are resampled, to keep the dependence over time; 2000 resamples (Seminar 3, problem B1)
  - bootstrap s.e. is 2.61%, so the interval is slightly wider than $\pm 1.96 \times 2.45\%$

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Betas Are Not Constant



- OLS beta on the last 252 daily log returns, recomputed every day
- Financials (XLF): from 0.60 to 2.17
- The peak follows the 2008–2009 banking crisis
- Technology (XLK): from 0.86 to 1.90
- High in the dot-com bubble (1998–2001), 1.75 at the end
- Energy (XLE): from −0.31 to 1.49
- Negative at the end, when oil moved against the market
- Utilities (XLU): from 0.06 to 1.00, the classic defensive sector
- A full-sample beta hides these regimes
- Conditional models: Chapters 5–6

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Shrinking Betas: Blume and Vasicek (1/2)

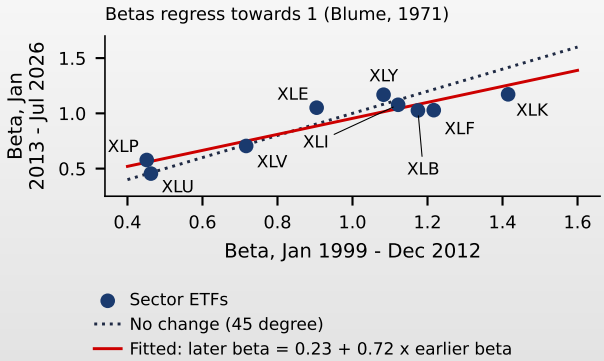
- **Regression to the mean**: a very high or very low estimated beta is partly noise, so in the next period it tends to move towards 1
- Blume (1971): regression of betas in one period on betas in the previous period
- Hence the practitioner rule: $\beta^{\text{Blume}} = 0.33 + 0.67\,\hat{\beta}$
- Vasicek (1973): a weighted average of the asset's estimate and the mean $\bar{\beta}$ of the betas of all assets (the prior)

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i)\bar{\beta}, \qquad w_i = \frac{\sigma_\beta^2}{\sigma_\beta^2 + \text{se}(\hat{\beta}_i)^2}$$

Shrinking Betas: Blume and Vasicek (2/2)

- w_i: the weight of the estimate, larger when $\hat{\beta}_i$ is precise (small s.e.)
- $\text{se}(\hat{\beta}_i)$: its standard error
- σ^2_β: the variance of the **true** betas across assets
- The variance of the estimates across assets, $\text{Var}_{cs}(\hat{\beta})$, adds true variance and noise: $\text{Var}_{cs}(\hat{\beta}) = \sigma^2_\beta + \overline{\text{se}^2}$, where $\overline{\text{se}^2}$ is the mean squared standard error
- So $\sigma^2_\beta = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$ (the empirical-Bayes estimate; derivation: Seminar 3, problem A3)
- Example: XLK on SPY, monthly 1999–2012 (on SPY, not Mkt-RF, hence a different beta from 1.28), with the prior estimated from the 9 sectors
  - $\hat{\beta} = 1.415$ (Newey–West s.e. 0.139), $\bar{\beta} = 0.949$
  - $\sigma^2_\beta = 0.116 - 0.008 = 0.108$, so $w = 0.1082/(0.1082 + 0.139^2) = 0.849$ (with unrounded inputs)
  - $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$
  - the Blume adjustment gives 1.278
  - the beta estimated later, on 2013–2026, is 1.172

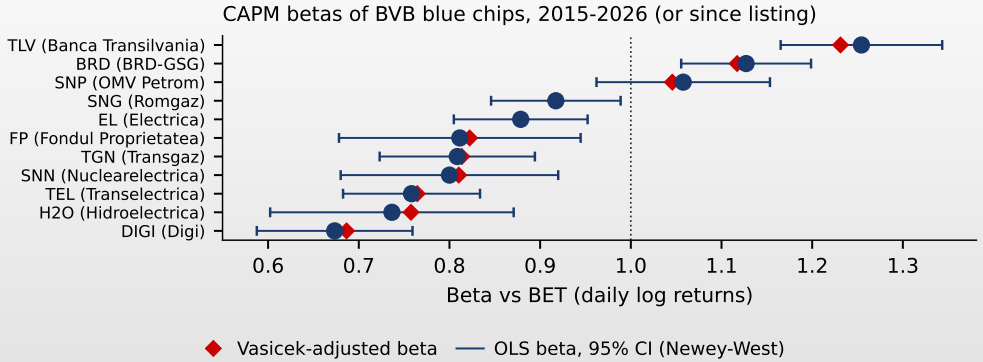
Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line $\hat{\beta}_{\text{later}} = 0.23 + 0.72\,\hat{\beta}_{\text{earlier}}$, very close to Blume's $0.33 + 0.67\,\hat{\beta}$
- **RMSE** (Root Mean Squared Error) of the forecast of later betas, $\sqrt{\frac{1}{N}\sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$: unadjusted beta 0.135, Blume 0.109, Vasicek 0.114
- Shrinkage cuts the forecast error by 15–20%
- With only 9 sectors the lesson is qualitative

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CAPM Betas on the Bucharest Stock Exchange (1/2)



Legend: OLS beta with its 95% confidence interval (CI), Newey–West errors; Vasicek-adjusted beta

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CAPM Betas on the Bucharest Stock Exchange (2/2)

- Daily log returns, regressed on BET, from 6 January 2015 or the listing date (Digi, DIGI, 2017; Hidroelectrica, H2O, 2023)
- Days with $|r_t| > 0.5$ are removed (one for DIGI, two for Fondul Proprietatea, FP)
- Banks are the most market-sensitive: Banca Transilvania (TLV) 1.25, BRD–Groupe Société Générale (BRD) 1.13
- OMV Petrom has a beta close to 1: 1.06
- Less sensitive: Transelectrica (0.76) and Digi (0.67)
- R^2: 0.59 for TLV, only 0.21 for DIGI
- A stock with a large weight in BET also has a high R^2 because it is itself part of the index
- Vasicek changes the estimates little, because daily betas are precise (s.e. between 0.04 and 0.07)
- Stale prices bias the daily betas of less liquid stocks (Seminar 3, problem B4)
- Caution: BET excludes dividends, while the stock prices are dividend-adjusted
  - alpha would be pushed up by roughly beta times the dividend yield of the index
  - beta depends on daily co-movement, so it should change little with BET-TR (the total-return version, dividends reinvested)

MFM_ch3_bvb_betas

Quick Check: A Beta, Its Error and Its Shrinkage

- A stock, 120 monthly returns: $\hat{\beta} = 0.90$, classical s.e. 0.05, Newey–West s.e. 0.09
  1. Give both 95% intervals. Is the stock less risky than the market under each?
  2. Which interval do you report, and why does the point estimate not change?
- A cross section of betas has $\mathrm{Var}_{cs}(\hat{\beta}) = 0.10$, $\overline{\mathrm{se}^2} = 0.02$ and mean $\bar{\beta} = 1.0$
  3. Compute σ_β^2 and the Vasicek beta of an asset with $\hat{\beta} = 1.6$, once with s.e. 0.2 and once with s.e. 0.1
  4. Which of the two is shrunk more, and why? Compare both with the Blume beta
- Answers: ▸ Appendix

From One Alpha to the GRS Test (1/2)

- **Jensen's alpha** (Jensen (1968)): the market-model intercept α_i
  - The CAPM implies $\alpha_i = 0$ for every asset
- A separate t-test for each asset: XLK has $t = 0.13$
  - With many assets, some alphas look significant by chance alone (Section 8)
- **The GRS test** (Gibbons–Ross–Shanken, Gibbons, Ross & Shanken (1989)): one test for all assets, $H_0 : \alpha_1 = \cdots = \alpha_N = 0$

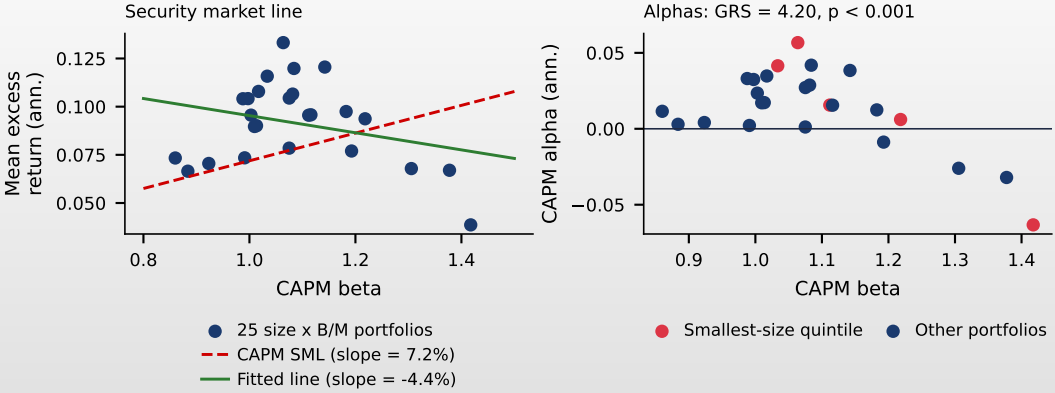
$$W = \frac{T - N - 1}{N} \, \frac{\hat{\boldsymbol{\alpha}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\alpha}}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \;\sim\; F(N,\, T - N - 1)$$

- Notation
  - N: number of test assets; T: number of months; $\hat{\boldsymbol{\alpha}}$: the N estimated alphas
  - $\hat{\boldsymbol{\Sigma}}$: maximum-likelihood residual covariance (divided by T)
  - $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market

From One Alpha to the GRS Test (2/2)

- $\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}$: a weighted sum of squared alphas; an alpha counts more when the asset's residuals are less volatile
- Decision: reject H_0 when W exceeds the 95% quantile of $F(N, T-N-1)$
- F: the Fisher distribution
- The distribution is exact, conditional on the factors
  - errors i.i.d. (independent and identically distributed) multivariate Normal, independent of the factors
  - zero mean and constant nonsingular covariance
- Interpretation: a large W means the market portfolio is far from the efficient frontier built from the market and the test assets

The Security Market Line in the Data (1/2)



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The Security Market Line in the Data (2/2)

- Test assets: US stocks sorted each June into 5 size and 5 B/M quintiles, 25 value-weighted portfolios (Fama & French (1993)), July 1963 – July 2026, $T = 757$
- The CAPM line has slope 7.2% a year (chart axes in annual fractions: $0.05 = 5\%$)
- The fitted line has slope -4.4%: higher beta, **lower** average return
- Betas span only 0.86–1.42, average excess returns 3.9%–13.3%
- Alpha is $+5.7\%$ a year for small firms with high B/M (small-value) and -6.3% a year for small firms with low B/M (small-growth)
- 10 of the 25 alphas have $|t| > 1.96$
- GRS $= 4.20$, p-value 8×10^{-11}: the CAPM is rejected

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GRS as a Sharpe-Ratio Test

- An algebraic identity (Gibbons, Ross & Shanken (1989)), with all moments estimated by maximum likelihood

$$\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}} = \widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)$$

  - $\widehat{SR}^2(f,\mathrm{R})$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets
  - $\widehat{SR}^2(f)$: that of the factor alone
- So GRS measures how much the Sharpe ratio rises when the test assets are added

$$W = \frac{T-N-1}{N}\cdot\frac{\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)}{1+\widehat{SR}^2(f)}$$

- On the 25 portfolios, the monthly Sharpe ratio of the market alone is 0.134 (unrounded 0.1344; times $\sqrt{12}$: 0.47 a year)
- The monthly Sharpe ratio of the tangency portfolio of the market and the 25 portfolios is 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1+0.134^2) = 4.20$, with $29.24 = (757-25-1)/25$
- If H_0 were true, noise alone would produce a much smaller gain (Appendix)
- Proof of the identity, and these numbers as its replication target: Seminar 3, problem A4

The Generalized Method of Moments in Brief (1/2) *(deeper)*

- An economic model implies $\mathbb{E}[\mathrm{g}_t(\boldsymbol{\theta})] = 0$ for a vector of moment conditions g_t
- **GMM** (Generalized Method of Moments, Hansen (1982)) chooses $\hat{\boldsymbol{\theta}}$ so that the sample mean of the moments is zero

$$\bar{\mathrm{g}}(\hat{\boldsymbol{\theta}}) = \tfrac{1}{T}\sum_{t=1}^{T} \mathrm{g}_t(\hat{\boldsymbol{\theta}}) = 0$$

  - exactly identified case: as many moments as parameters
  - with more moments than parameters, not all can be zero, and $\bar{\mathrm{g}}$ is made as small as possible with a weighting matrix
- Sandwich variance in the exactly identified case

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) \approx \tfrac{1}{T}\,\mathrm{D}^{-1}\mathrm{S}\,\mathrm{D}^{-1\top}, \qquad \mathrm{D} = \mathbb{E}\big[\partial \mathrm{g}_t/\partial \boldsymbol{\theta}^{\top}\big], \qquad \mathrm{S} = \sum_{j=-\infty}^{\infty} \mathrm{Cov}(\mathrm{g}_t, \mathrm{g}_{t-j})$$

  - D: Jacobian of the moments (their derivatives with respect to the parameters)
  - S: long-run variance of the moments, autocovariances included, estimated by Newey–West (Section 3)
- The simplest case, a mean: $g_t = R_t - \mu$ gives $D = -1$ and $\mathrm{Var}(\hat{\mu}) \approx S/T$; this is the HAC t-statistic of a mean factor return (Do the Premia Survive?)
- OLS: $\mathrm{g}_t = \mathrm{x}_t(y_t - \mathrm{x}_t^{\top}\boldsymbol{\theta})$ gives $\mathrm{D} = -\mathbb{E}[\mathrm{x}_t\mathrm{x}_t^{\top}]$: White errors when S has no lags, Newey–West errors otherwise (XLK table)

The Generalized Method of Moments in Brief (2/2) *(deeper)*

- **GRS as GMM**: parameters (α_i, β_i) for $i = 1, \dots, N$; $2N$ moments $\varepsilon_t \otimes (1, f_t)$, each residual times 1 and times the factor ($\otimes$: Kronecker product)
  - the system is exactly identified, so the estimates are the OLS ones, asset by asset; GMM changes only their variance
  - ordering the parameters asset by asset, $(\alpha_1, \beta_1, \alpha_2, \beta_2, \dots)$, gives $\mathrm{D} = -\mathrm{I}_N \otimes (\mathrm{Z}^\top \mathrm{Z}/T)$, with Z the $T \times 2$ matrix of rows $(1, f_t)$
  - Seminar 3, problem B7 puts all the alphas first; the resulting test is the same
- $\hat{\mathrm{V}}_\alpha$: the block of the sandwich matrix that belongs to the alphas, computed with a HAC $\hat{\mathrm{S}}$
- Wald statistic $W_{\mathrm{GMM}} = \hat{\boldsymbol{\alpha}}^\top \hat{\mathrm{V}}_\alpha^{-1} \hat{\boldsymbol{\alpha}} \to \chi^2_N$ under H_0
- The same machinery gives the misspecification-robust premia errors (Appendix) and estimates the discount-factor model of Section 7

When the GRS Assumptions Fail

- **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
  - Residual bootstrap under H_0: resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar 3, problem B7)
  - The bootstrap is robust to non-Normality only if the errors are i.i.d. and independent of the factors
  - GMM Wald test: moments $\varepsilon_t \otimes (1, f_t)$, each residual times 1 and times the factor ($\otimes$: Kronecker product; set-up and ordering: GMM in Brief)
  - $W_{\text{GMM}} = \hat{\boldsymbol{\alpha}}^\top \hat{\mathrm{V}}_\alpha^{-1} \hat{\boldsymbol{\alpha}} \to \chi^2_N$, $\hat{\mathrm{V}}_\alpha$: HAC estimate of $\mathrm{Var}(\hat{\boldsymbol{\alpha}})$
  - robust to heteroskedasticity and autocorrelation
- **Too many assets for the number of months**: $\hat{\boldsymbol{\Sigma}}$ has rank at most $T - 2$
  - when $N \geq T - 1$, $\hat{\boldsymbol{\Sigma}}$ cannot be inverted and GRS cannot be computed
  - the power of the test falls well before that limit
- Tests built for many assets: Appendix

A Flat SML: Black and Betting Against Beta



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): if investors cannot borrow freely at R_f, expected return stays linear in beta, but the line starts at the return of a **zero-beta** portfolio, which is above R_f
- Prediction: the SML is **flatter** than the CAPM line
- Frazzini & Pedersen (2014): investors who want more risk but cannot borrow buy high-beta assets directly
- The extra demand makes these assets expensive, so their future returns are lower
- Their strategy, **BAB** (Betting Against Beta)

$$r_t^{\mathbf{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L: low-beta portfolio, leveraged up to beta 1
- H: high-beta portfolio, scaled down to beta 1
- Concretely, per unit of capital: $1/\beta_L > 1$ units go into L (the difference is borrowed) and $1/\beta_H < 1$ units into H (the rest is lent at R_f)
- BAB has beta zero by construction and a positive average return in their data
- A plain low-minus-high-beta spread, without scaling, has a negative beta: its mean mixes the BAB premium with a negative market exposure (Seminar 3, problem C2)

What a CAPM Test Establishes

- Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth
- Stock indices omit bonds, real estate, private firms and human capital
- Mathematically, betas computed against **any** efficient portfolio price the mean returns of all assets exactly (Seminar 3, problem A2)
  - so a CAPM test really checks whether the index used as the market is efficient
  - a rejection can come from the model or from the choice of index
- **Test assets matter**: 9 sector ETFs, January 1999 – July 2026 ($T = 331$): GRS $= 0.53$, $p = 0.85$, not rejected
  - These are the sectors of the frontier slide, with Mkt-RF as the market (Sharpe ratio 0.51 a year, close to the 0.49 of SPY)
  - Adding the sectors raises the squared monthly Sharpe ratio by 0.015, less than the 0.029 that noise alone would produce (Appendix)
  - On the same 331 months the 25 size $\times$ B/M portfolios still reject: GRS $= 2.91$, $p = 9 \times 10^{-6}$
  - The reason: each sector mixes small and big, value and growth firms, so the sectors differ little in the characteristics that explain returns
- Conclusion: failing to reject a model is weak evidence in its favour
- Test models on assets whose mean returns differ widely

Quick Check: Reading a GRS Statistic

- One factor, $N = 4$ test assets, $T = 65$ months; monthly Sharpe ratios: factor alone 0.15, tangency portfolio of the factor and the 4 assets 0.25
  1. Compute W from the Sharpe-ratio form. The 95% quantile of $F(4, 60)$ is about 2.53: reject?
  2. Compute the squared Sharpe-ratio gain expected under H_0, $N(1 + \widehat{SR}^2(f))/(T - N - 3)$, and compare it with the observed gain
  3. Same Sharpe ratios, $T = 650$ months: recompute W (95% quantile of $F(4, 645)$: about 2.39). What changed, the gap or the evidence?
- Link to the chapter: the sector frontier shows a visible gap to SPY, yet GRS does not reject on 331 months
- Answers: Appendix

Arbitrage Pricing Theory

- **APT**: Arbitrage Pricing Theory, Ross (1976)
- Assumption: returns follow a K-factor model, and asset-specific risks can be diversified away

$$R_{i,t} = \mathbb{E}[R_i] + \boldsymbol{\beta}_i^\top \mathsf{f}_t + \varepsilon_{i,t}, \qquad \mathbb{E}[\mathsf{f}_t] = 0$$

- **No arbitrage** (no sure profit without capital invested) implies, approximately

$$\mathbb{E}[R_i] - R_f = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$$

- $\boldsymbol{\beta}_i$: the K loadings of asset i
- f_t: the unexpected part of the factors, with mean 0
- $\boldsymbol{\lambda}$: the **risk premia**, the extra expected return per unit of each exposure
- The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

size: NYSE median; B/M: 30% / 40% / 30%

	Low (growth)	Neutral	High (value)
Small	S_L	S_N	S_H
Big	B_L	B_N	B_H

$$\text{SMB} = \tfrac{1}{3}(S_L + S_N + S_H) - \tfrac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \tfrac{1}{2}(S_H + B_H) - \tfrac{1}{2}(S_L + B_L)$$

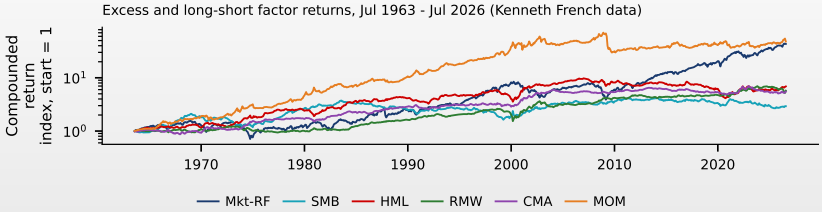
- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- The short side finances the long side, so the factor return is a difference of returns, $f_t = R_{\textbf{high},t} - R_{\textbf{low},t}$
- Fama–French: each June, stocks are split into 2×3 groups using the previous year's accounting data; each group is a value-weighted portfolio
- The size breakpoint is the median market capitalisation of NYSE (New York Stock Exchange) stocks
  - **SMB** (Small Minus Big): small minus big stocks
  - **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- The same construction on profitability and investment gives RMW and CMA (next slide)

The Standard Factor Models

Model	Factors	Reference	Motivation
CAPM	Mkt-RF	Sharpe (1964)	equilibrium
FF3	Mkt-RF, SMB, HML	Fama & French (1993)	size and B/M effects (Fama & French (1992))
Carhart	FF3 + MOM	Carhart (1997)	fund evaluation
q-factor	market, size, investment, ROE	Hou, Xue & Zhang (2015)	firms' investment decisions
FF5	Mkt-RF, SMB, HML, RMW, CMA	Fama & French (2015)	dividend-discount model

- FF3 and FF5: the Fama–French three- and five-factor models; the FF3 regression is $R^e_{i,t} = \alpha_i + b_i\,\text{MKT}_t + s_i\,\text{SMB}_t + h_i\,\text{HML}_t + \varepsilon_{i,t}$, with MKT = Mkt-RF and loadings b_i, s_i, h_i
- A good model leaves time-series alphas close to zero, so GRS does not reject it
- Factors beyond FF3 (ROE: Return On Equity)
  - **RMW** (Robust Minus Weak): high minus low operating profitability
  - **CMA** (Conservative Minus Aggressive): low minus high asset growth
  - **MOM** (momentum, also UMD, Up Minus Down): stocks with high minus stocks with low returns over months $t-12$ to $t-2$ (the last month is skipped)
  - Past winners keep winning for 3–12 months (Jegadeesh & Titman (1993))
- Note: the FF3 SMB comes from one sort (size and B/M), while the FF5 SMB averages three sorts
- For this reason FF3 is not a special case of FF5

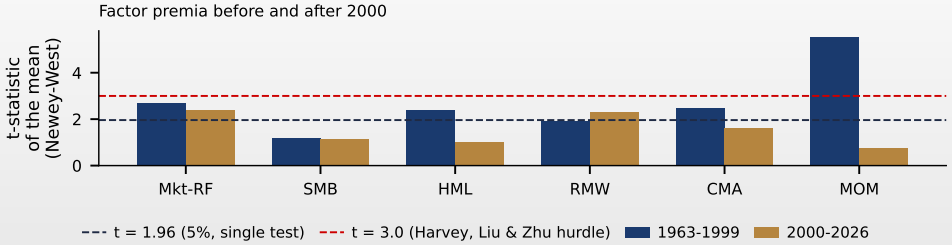
Six Factors over Sixty-Three Years



- Monthly factor returns, compounded from 1, on a log scale
- The factors are excess or long–short returns, so the curves include no interest on capital or on the collateral of short positions
- The curves are therefore not the wealth of a real investor
- Final value of one unit invested, July 1963 – July 2026: Mkt-RF 43.2, MOM 48.0, HML 6.9, RMW 5.7, CMA 5.5, SMB 2.9
- Momentum peaked before 2009 and lost heavily when the market rebounded sharply in 2009: a **momentum crash**
- The value factor was flat or losing from about 2007 to 2020
- Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

MFM_ch3_factor_cum

Do the Premia Survive?



- t-statistic of the mean monthly factor return (Newey–West), 1963–1999 vs 2000–2026
- Momentum: $t = 5.53$ before 2000 and 0.75 after
- The annual momentum mean fell from 10.7% to 2.5%
- Value (HML): $t = 2.39$ before 2000 and 1.02 after
- The annual HML mean fell from 4.2% to 2.8%
- Profitability (RMW) is the exception: $t = 1.90$, then 2.31
- The market premium stays significant in both halves ($t = 2.70$ and 2.39)
- The year 2000 is an arbitrary cut, not the publication date of the factors; the decay linked to publication is tested in Section 12

MFM_ch3_factor_tstats

The Two Passes



Eugene Fama, Morgan Stanley–AFA (American Finance Association) Award, 2008

Photo: TonyTheTiger (2008); CC BY-SA 3.0; Wikimedia Commons

- Fama & MacBeth (1973) estimate the risk premia $\boldsymbol{\lambda}$ of $\mathbb{E}[R_i^e] = \lambda_{\mathbf{0}} + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ (Fama: Nobel Prize 2013)
- **Pass 1**: time series, one regression per asset
  - returns $T \times N$, factors $T \times K \Rightarrow$ betas $\hat{\mathsf{B}}$, $N \times K$

$$R_{i,t}^e = a_i + \boldsymbol{\beta}_i^\top \mathsf{f}_t + \varepsilon_{i,t}$$

- **Pass 2**: cross section, each month one regression of the N returns on $\mathsf{X} = [1, \hat{\mathsf{B}}]$

$$R_{i,t}^e = \lambda_{\mathbf{0},t} + \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \qquad \hat{\boldsymbol{\lambda}} = \tfrac{1}{T} \textstyle\sum_t \hat{\boldsymbol{\lambda}}_t$$

- $\lambda_{\mathbf{0}}$: the intercept, zero if the model is right (in Black (1972): how far the zero-beta return exceeds R_f)
- a_i: pass-1 intercept
- $u_{i,t}$: pass-2 residual
- $\hat{\boldsymbol{\lambda}}$: average of the monthly slopes
- The method also works with factors that are **not** traded portfolios (for example, inflation) and with firm characteristics as regressors

Standard Errors for Risk Premia (1/2)

- **Fama–MacBeth**: the monthly $\hat{\lambda}_{k,t}$ form a sample of size T

$$\text{se}_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$$

  - $s(\cdot)$: the sample standard deviation over the T months
  - robust to correlation of the residuals across assets in the same month
  - if $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series
- **Misspecified model**: the Fama–MacBeth, Newey–West and Shanken (next slide) errors assume that the model is correct
  - that is, the pricing errors are zero: $\mathrm{e} = \boldsymbol{\mu} - \mathrm{X}\boldsymbol{\gamma} = 0$
  - $\boldsymbol{\mu}$: mean excess returns, $\boldsymbol{\gamma} = (\lambda_0, \boldsymbol{\lambda}^\top)^\top$
  - if the model is wrong ($\mathrm{e} \neq 0$), the noise in the estimated betas adds variance to the premia (Kan, Robotti & Shanken (2013); formula: Appendix)

Standard Errors for Risk Premia (2/2)

- **Errors in variables**: pass 2 uses estimated betas, not the true ones, so se_{FM} understates the uncertainty
- The Shanken (1992) correction scales by $(1 + c)$ the part of the variance that does not come from factor variability

$$\text{Var}_{\text{Sh}}(\hat{\boldsymbol{\lambda}}) = (1 + c)\Big[\text{Var}_{FM}(\hat{\boldsymbol{\lambda}}) - \tfrac{1}{T}\boldsymbol{\Sigma}_f\Big] + \tfrac{1}{T}\boldsymbol{\Sigma}_f, \qquad c = \hat{\boldsymbol{\lambda}}^\top \boldsymbol{\Sigma}_f^{-1}\hat{\boldsymbol{\lambda}}$$

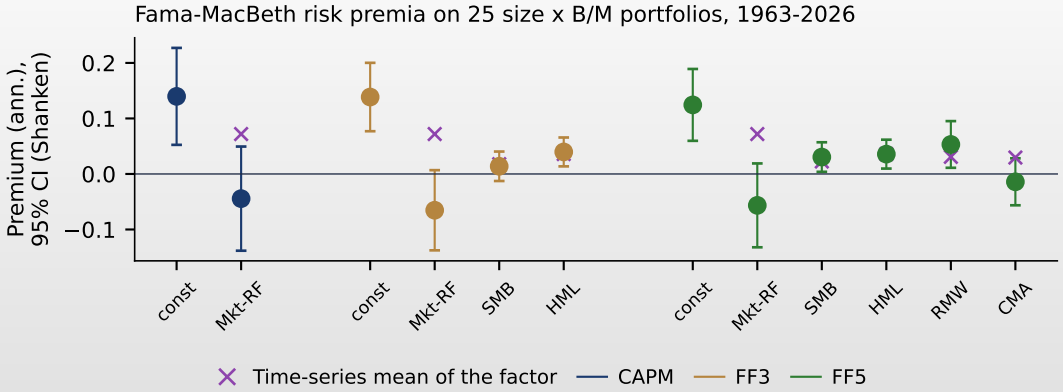
  - Var_{FM}: the Fama–MacBeth variance, $s^2(\hat{\boldsymbol{\lambda}}_t)/T$
  - $\boldsymbol{\Sigma}_f$: factor covariance
  - c: squared Sharpe ratio of the premia, per month; 0.03 for FF3 here, but possibly much larger for non-traded factors
- Small c (traded factors, monthly data): the correction is small
- Large c: Fama–MacBeth t-statistics are overstated

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	**s.e.**	t	**Significant at 5%?**
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- 25 size $\times$ B/M portfolios, 1963–2026
- The Shanken correction barely moves the error, because $c = 0.03$ per month
- What matters most is the autocorrelation of the monthly slopes $\hat{\lambda}_t$, corrected by Newey–West with 6 lags ($T = 757$)
- Kan–Robotti–Shanken: the error robust to a wrong model; for RMW in FF5 it rises from 2.1% to 2.6%, and t falls from 2.5 to 2.1
- HML clears the usual 1.96 hurdle with all four errors
- The stricter $t > 3$ hurdle (Section 8) is only just reached with Kan–Robotti–Shanken (3.00) and clearly missed with Newey–West (2.45)

Estimated Risk Premia (1/2)



MFM_ch3_fama_macbeth

Estimated Risk Premia (2/2)

- 25 size $\times$ B/M portfolios, 1963–2026
- Bars: 95% intervals with Shanken errors, in annual fractions ($0.1 = 10\%$)
- Crosses: factor means
- CAPM: intercept 14.0% (s.e. 4.4%), market premium -4.4% (s.e. 4.8%)
- Estimated premia, standard error in brackets
  - FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%)
  - FF5: RMW 5.3% (2.1%), CMA -1.4% (2.2%)
- Traded factor: λ should equal its time-series mean
- HML: estimated premium 4.0%, factor mean 3.6%, so the two are close
- Market (FF3): estimated premium -6.5%, factor mean 7.2%, so the two are far apart
- Why? Market betas differ little across the portfolios: 0.86–1.42 in the CAPM and 0.89–1.14 in FF3
- With so little spread, the market premium is **weakly identified**: the intercept absorbs the average level of returns (Lewellen, Nagel & Shanken (2010))

MFM_ch3_fama_macbeth

Useless and Weak Factors *(deeper)*

- A **useless factor** g_t is independent of all returns: its true betas are zero, and the estimated ones are pure noise, of order $1/\sqrt{T}$
- Pass 2 regresses returns on this noise: $\hat{\lambda}_g$ does not go to zero, and the t-test too often declares the factor priced when the model is wrong (Kan & Zhang (1999))
- Simulation on the 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
  - $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
  - Remedy: first test in pass 1 whether all the factor's betas are zero (Wald test); this test rejects in only 6.7% of the replications, close to the nominal level
- **Weak identification**: betas close to zero (a weak factor) or nearly equal across assets (the market, in our example)
  - identification-robust tests (Kleibergen (2009))
  - with many assets, consistency of pass 2 depends on factor strength (Anatolyev & Mikusheva (2022))

A Skeptical Appraisal: Adding 30 Industry Portfolios *(deeper)*

	OLS R^2		GLS R^2	
Model	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

Monthly 1963–2026, free intercept and premia; replication target of Seminar 3, problem B6

- The returns of the 25 portfolios are almost fully explained by three factors: any factors correlated with SMB and HML reproduce their means
- So a high OLS R^2 on these portfolios proves little (Lewellen, Nagel & Shanken (2010))

A Skeptical Appraisal: Three Checks *(deeper)*

- Three checks (Lewellen, Nagel & Shanken (2010))
  - Broaden the test assets: FF3 falls from 0.66 to 0.22 once 30 industries are added
  - Also report the **GLS** (Generalised Least Squares) R^2, which weights the errors by the inverse covariance of returns, like GRS: never above 0.26
  - Force the premium of each traded factor to equal its mean and drop the intercept: FF3 on the 55 assets gives $R^2 = -0.27$
  - $R^2 = 1 - \text{SSE}/\text{SST}$, with SSE (Sum of Squared Errors) and SST (Total Sum of Squares, deviations from the mean)
  - R^2 is negative when the model does worse than the plain average of the returns; OLS with an intercept cannot do worse than that average
- Individual stocks instead of portfolios: betas are more dispersed, but also estimated with more noise (Ang, Liu & Schwarz (2020))

Quick Check: Premia, Shanken and a Negative R^2

- A non-traded factor (for example, industrial-production growth) with volatility $\sigma_f = 1\%$ a month; $T = 400$ months; pass 2 gives $\hat{\lambda} = 0.5\%$ a month, with $s(\hat{\lambda}_t) = 3\%$
  1. Compute se_{FM} and the Fama–MacBeth t-statistic
  2. Compute c, the Shanken variance and the Shanken t-statistic. Does the factor clear the $t > 3$ hurdle?
  3. Why would c be tiny for a traded factor with $\sigma_f = 4.5\%$ and the same $\hat{\lambda}$?
- Three portfolios have mean excess returns 2%, 4%, 6% a year; a model with imposed premia and no intercept fits 6%, 4%, 2%
  4. Compute SST, SSE and R^2
  5. Why can the R^2 of an OLS regression with an intercept never be negative?
- Answers: ▸ Appendix

Factors as a Stochastic Discount Factor *(deeper)*



Lars Peter Hansen, Nobel Prize in Economics 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- **SDF** (Stochastic Discount Factor) m_t: prices all excess returns, $\mathbb{E}[m_t R^e_{i,t}] = 0$
- m_t is high in bad times (for example, when the market falls); an asset that pays off exactly then works as insurance, so investors accept a low average return on it
- A linear factor model is an SDF, $m_t = 1 - \mathrm{b}^\top(\mathrm{f}_t - \mathbb{E}\mathrm{f})$
- b: the SDF coefficients, $\mathbb{E}$f: the factor means
- Substitute: $\mathbb{E}[R^e_i] = \mathrm{Cov}(R^e_i, \mathrm{f}_t)^\top \mathrm{b} = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ with $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathrm{b}$: the beta pricing model of Section 6
- Testing $\mathrm{b} = 0$ asks whether a factor helps price the assets, given the other factors
- Testing $\boldsymbol{\lambda} = 0$ asks whether exposure to the factor is rewarded
- These moment conditions are estimated by GMM, Hansen (1982) (GMM in Brief: here $\mathrm{g}_t(\mathrm{b}) = m_t(\mathrm{b})\, \mathrm{R}^e_t$)
- A survey of this view: Cochrane (2011)

Comparing Models: Maximum Sharpe Ratios *(deeper)*

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): the better model is the one whose factors allow the highest Sharpe ratio, $SR^2(\mathrm{f}) = \boldsymbol{\mu}_f^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f$ ($\boldsymbol{\mu}_f$, $\boldsymbol{\Sigma}_f$: means and covariance of the factors)
- The comparison does not depend on the choice of test assets
- When one model contains the other, in-sample SR^2 can only rise
- Then run a **spanning regression**: regress the added factor on the others; a non-zero alpha shows that it brings something new
  - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$)
  - since 2000: $\alpha = -1.7\%$ ($t = -0.81$), so no significant incremental contribution of HML, as in Fama & French (2015)
- FF5 vs Carhart (neither contains the other): the difference in annual SR^2 is 0.00, with 95% interval $[-0.57, 0.66]$
- Since 2000: the difference is 0.50, with 95% interval $[-0.24, 1.48]$, so again not significant (stationary bootstrap, Politis & Romano (1994); Seminar 3, problem B8)
- Another metric common to all models, the Hansen–Jagannathan distance: ▸ Appendix

The Factor Zoo

- Cochrane (2011) called the growing list of new factors a "zoo"
- Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- Each factor was tested as if it were the only one tried: $|t| > 1.96$ at 5%
- With many tests, some factors pass by chance
- **Data mining** (or p-hacking): searching across many variants until one looks significant
- Their proposal: a new factor needs $|t| > 3.0$

Controlling False Discoveries

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- **FWER** (Family-Wise Error Rate): probability of at least one false rejection (Chapter 2)
- Notation of the table: M: number of tests
- α: significance level (not a pricing error)
- $p_{(1)} \leq \cdots \leq p_{(M)}$: sorted p-values
- $H_{(j)}$: the hypothesis with the j-th smallest p-value
- **BH**: the Benjamini–Hochberg procedure; **BY**: the Benjamini–Yekutieli procedure

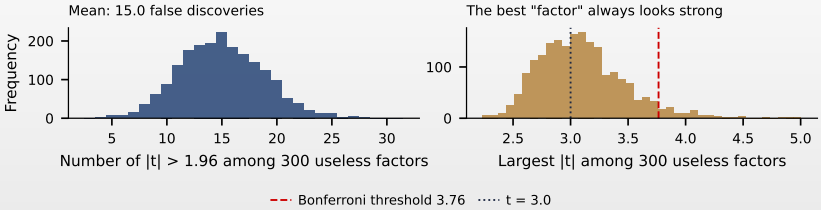
Procedure	Controls	Reject $H_{(j)}$ if	Valid under
Bonferroni	FWER	$p_{(j)} \leq \alpha/M$	any dependence
Holm (Holm (1979))	FWER	$p_{(k)} \leq \alpha/(M-k+1)$ for all $k \leq j$	any dependence
BH (Benjamini & Hochberg (1995))	FDR	$j \leq$ largest k with $p_{(k)} \leq k\alpha/M$	positive dependence
BY (Benjamini & Yekutieli (2001))	FDR	as BH, thresholds divided by $\sum_{k=1}^{M} 1/k$	any dependence

- Positive dependence: the test statistics tend to move up together; independence is a special case (the technical condition is called PRDS, positive regression dependence on a subset, Benjamini & Yekutieli (2001))
- FWER: suitable when no false discovery is acceptable
- FDR: suitable when searching over many candidate factors and accepting a small share of false discoveries

Worked Example: Ten Candidate Factors

- Sorted p-values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62
- Significance level $\alpha = 5\%$
- Naive ($p \leq 0.05$): **7** discoveries
- Bonferroni ($p \leq 0.005$): **2**
- Holm compares the p-values in turn with $0.05/10$, $0.05/9$, $0.05/8$, ...: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, then $0.011 > 0.0071$ and it stops: **3**
- BH compares $p_{(j)}$ with $0.005j$; the last p-value under its threshold is $p_{(6)} = 0.027 \leq 0.030$: **6**
- Seminar 3, problem A1 repeats the procedure on a different vector, with a threshold chart

Simulation: 300 Useless Factors



- 2000 replications; each draws 300 independent t-statistics from the standard Normal distribution (no true premium)
- On average 15.0 useless factors pass $|t| > 1.96$, as expected: $300 \times 0.05 = 15$
- On average 0.83 useless factors pass $|t| > 3$
- At least one useless factor passes this hurdle in 57% of the replications
- The median of the largest $|t|$ in a replication is 3.06
- Bonferroni threshold for 300 tests: $|t| > 3.76$
- The right hurdle depends on how many tests were actually tried, unpublished ones included

MFM_ch3_false_discoveries

Quick Check: Counting Discoveries

- Five candidate factors, sorted p-values 0.004, 0.012, 0.020, 0.030, 0.200; significance level $\alpha = 5\%$
  1. How many discoveries under the naive rule, Bonferroni, Holm and BH? Write down each threshold you compare with
  2. Under BY the BH thresholds are divided by $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$: how many discoveries remain?
  3. If all five nulls were true and the tests independent, what is the probability that at least one p-value falls below 0.05?
- Which count would you report for a search over many candidate factors?
- Which count would you report for a claim that **no** factor found is spurious?
- Answers: Appendix

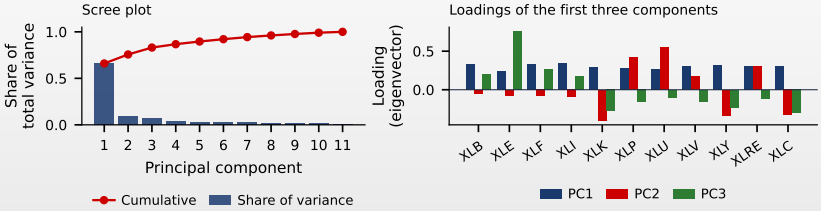
Principal Components in Brief

- **Latent factors**: unobserved factors, estimated from returns alone, without saying in advance what they represent
- **PCA** (Principal Component Analysis) of the correlation matrix C of standardised returns

$$\mathrm{C} = \mathbf{\Gamma} \mathrm{D} \mathbf{\Gamma}^\top, \qquad \mathrm{y}_t = \mathbf{\Gamma}^\top \mathrm{z}_t$$

  - z_t: standardised returns of day t
  - y_t: the components
  - $\mathbf{\Gamma}$: eigenvectors in columns
  - D: diagonal matrix of the eigenvalues ℓ_k
- Eigenvalues, sorted: $\ell_1 \geq \cdots \geq \ell_N$ (written ℓ so as not to confuse them with the risk premia λ)
- The eigenvectors are the **loadings**
- Component k explains ℓ_k / N of the total variance
- Example, two assets with correlation ρ: $\ell = 1 \pm \rho$; the vector $(1,1)/\sqrt{2}$ is the "market", and $(1,-1)/\sqrt{2}$ is the difference between the two
- In a crisis ρ rises and the first component dominates
- Textbook: Härdle & Simar (2019), chapter on principal components
- N equicorrelated assets: Seminar 3, problem A8

PCA of the US Sector ETFs



- 11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)
- The first principal component (PC1) explains 66.0% of the variance
- All PC1 loadings are positive, between 0.24 and 0.34
- The correlation of PC1 with SPY is 0.954: the first component is, in practice, the market factor
- The second component, PC2 (9.7%): utilities $+0.55$, staples $+0.42$, real estate $+0.31$ vs technology -0.40, consumer discretionary -0.33: **defensive versus growth sectors**, although PCA was given no labels
- The third component, PC3 (7.4%), is dominated by energy
- The first three components explain 83.1%

MFM_ch3_pca

How Many Factors? *(deeper)*

- Approximate factor model $x_{it} = \boldsymbol{\Lambda}_i^\top \mathrm{F}_t + e_{it}$, with both the number of assets N and of periods T large
  - x_{it}: standardised return; F_t: factors; $\boldsymbol{\Lambda}_i$: loadings; e_{it}: idiosyncratic part
  - $V(k)$: mean squared residual after the first k components; ℓ_k: the k-th largest eigenvalue
- Bai & Ng (2002): choose the k that minimises $\ln V(k) + k\,\frac{N+T}{NT}\ln\min(N,T)$ (the information criterion IC_{p2}: fit plus a penalty per factor)
- Ahn & Horenstein (2013): choose the k where the ratio ℓ_k/ℓ_{k+1} is largest, that is, where the eigenvalues drop most sharply
- The rule of Kaiser (1960) ($\ell_k > 1$) has no theoretical justification: even pure noise produces eigenvalues scattered around 1

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size × B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns, with at most $k = 5$ factors for sectors and $k = 8$ for portfolios (5 portfolios with missing months are dropped)
- Ahn–Horenstein picks only the market factor (ratio 14.6); the second peak is at $k = 3$ (ratio 3.5), as in FF3
- Bai–Ng on 11 sectors picks the largest number allowed, $k = 5$: its penalty is designed for many assets, so with only 11 the answer is not reliable (Appendix)

Limits of PCA for Asset Pricing

- Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- Components explain **variance**, not necessarily **average returns**
- Individual stocks enter and leave the sample (listings, delistings), so many observations are missing, while classic PCA needs a complete panel
- The sign of each component is arbitrary: $-\gamma_k$ is as good as γ_k
- Latent factors are determined only up to a rotation: any rotated combination of them fits the data equally well, so interpreting them is a choice
- Two extensions of PCA also target average returns: risk-premium PCA and the three-pass estimator (▸ Appendix)
- PCA and factor analysis in finance: Franke, Härdle & Hafner (2019)

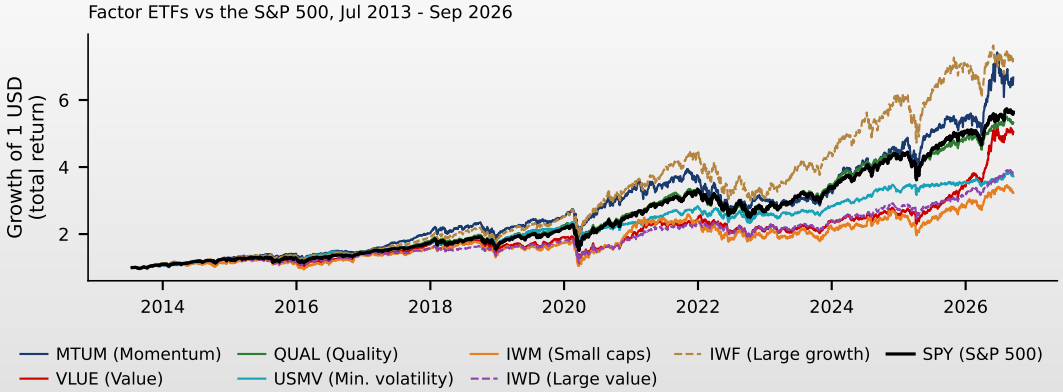
Instrumented PCA *(deeper)*

- **IPCA**: Instrumented Principal Component Analysis, Kelly, Pruitt & Su (2019)
- Loadings are functions of L observed characteristics $z_{i,t}$ (size, B/M, momentum, ...)

$$R^e_{i,t+1} = \alpha_{i,t} + \boldsymbol{\beta}^\top_{i,t} f_{t+1} + \varepsilon_{i,t+1}, \qquad \boldsymbol{\beta}^\top_{i,t} = z^\top_{i,t} \boldsymbol{\Gamma}_\beta$$

- $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors f_t are estimated jointly, in turns: given the factors, estimate $\boldsymbol{\Gamma}_\beta$, then the reverse
- The question tested: do characteristics predict returns because they measure risk exposure (through beta), or directly, as alpha ($\alpha_{i,t} = z^\top_{i,t} \boldsymbol{\Gamma}_\alpha$)?
- Result on US stocks: a few IPCA factors explain average returns better than models with as many observed factors
- $\boldsymbol{\Gamma}_\alpha$ is not significant, hence the paper's title: "characteristics are covariances"
- Crypto-asset factors: Appendix

Factor ETFs versus the Market (1/2)



MFM_ch3_factor_etfs

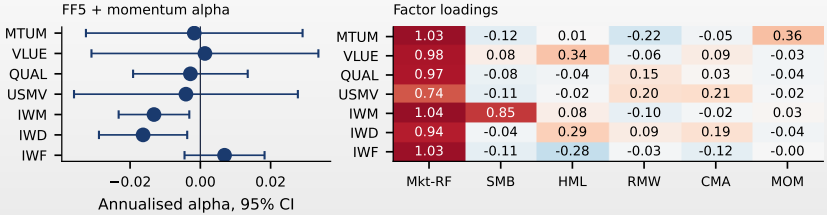
Factor ETFs versus the Market (2/2)

- **Smart beta**: ETFs that buy stocks by fixed rules to increase exposure to a factor; they sell nothing short (long-only)
  - so they keep market exposure and do not exactly replicate the academic long–short factor
- Thirteen years are a single historical path, too short to reject the factor premia

- **CAGR** (Compound Annual Growth Rate), $(V_T/V_0)^{1/n} - 1$ over n years, of 1 USD (US dollar) invested, dividends re-invested, from 18 July 2013
  - above SPY (14.0%): large growth (IWF) 16.3%, momentum (MTUM) 15.5%
  - below SPY: quality (QUAL) 13.5%, value (VLUE) 13.1%, large value (IWD) 10.7%, minimum volatility (USMV) 10.5%, small caps (IWM, Russell 2000 index) 9.3%
  - Russell indices: the Russell 1000 covers the 1000 largest US stocks, with value (IWD) and growth (IWF) sub-indices; the Russell 2000 covers the next 2000, smaller firms (IWM)

MFM_ch3_factor_etfs

Alphas of Factor ETFs



- Daily regressions on the six factors (FF5 + MOM), Newey–West errors, each ETF on its own trading days, 2000–2026
- In the chart, alphas are annual fractions ($0.02 = 2\%$)
- The loadings match the labels: MTUM on MOM 0.36, VLUE on HML 0.34, IWM on SMB 0.85
- IWF (growth) loads -0.28 on HML
- USMV (minimum volatility) has a market beta of 0.74
- Alphas close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%
- The Russell ETFs have negative alphas: IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.54$), explained by fees and by the index construction rules
- R^2 between 0.81 and 0.97: factor ETFs mostly sell exposure to known factors (Seminar 3, problem B11)

MFM_ch3_etf_alphas

Timing, Crowding and Costs

- **Factor timing**: moving capital between factors based on signals, for example how cheap a factor is or how it has done recently
  - Premia are uncertain and histories short, so it is hard to show that timing works out of sample (Chapter 8)
- **Crowding**: many investors hold the same factor positions
  - Premia can shrink (McLean & Pontiff (2016)), and crowding can amplify losses during forced liquidations
  - The 2009 momentum crash coincided with a sharp rebound of past losers (Daniel & Moskowitz (2016))
- **Implementability**: academic factor returns do not deduct trading costs or the cost of borrowing the shorted stocks
- An ETF pays these costs, so its return is lower

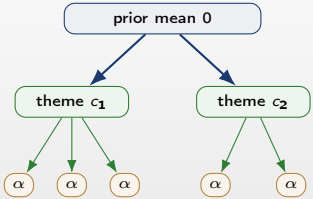
Do Published Factors Replicate?

- **Two critiques** of factor research (Jensen, Kelly & Pedersen (2023), Introduction)
  - Results cannot be reproduced: Hou, Xue & Zhang (2020) replicate only 35% of the anomalies they study
  - Results are the product of testing many factors without correction: Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Anomaly**: an average return that the CAPM does not explain
- **Questions**
  - do published factors have significant CAPM alphas in the original data?
  - are these alphas significant out of sample and in other countries?
  - do they remain significant once multiple testing is taken into account?

Why Replication Matters for This Chapter

- **Why it matters**
  - if most factors are false discoveries, multifactor models and factor investing (Sections 5 and 10) rest on noise
- **A second view of the zoo** (Feng, Giglio & Xiu (2020), Appendix)
  - most new factors add nothing to the existing ones
  - Jensen, Kelly & Pedersen instead see the zoo as many correlated measurements of the same few economic themes
- **Link to this chapter**
  - Jensen's alpha (Section 4) is the object tested
  - the Benjamini–Yekutieli correction (Section 8) is the classical (non-Bayesian) benchmark
  - Bayesian shrinkage applies to alphas the same idea as Vasicek for beta (Section 3)

153 Factors Grouped into 13 Themes *(deeper)*



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II): 153 factors in 93 countries
  - stocks are split into thirds by each characteristic, weighted by market capitalisation, with a cap
  - the factor buys the third with the higher return in the original study and sells the other extreme third
  - each factor is scaled to a specific volatility (uncorrelated with the market) of 10% a year, so that alphas are comparable
- **Themes**: 13 clusters of US factors
  - hierarchical clustering: factors whose CAPM residuals are strongly correlated fall into the same theme
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
  - $f_{i,t}$: return of factor i; r_t^m: market excess return
  - each factor's alpha is a sum: $\alpha_i = c_j + s_n + w_i$ (theme, signal and factor components), with prior mean 0

Shrinking Alphas and the Replication Rate *(deeper)*

- The single-factor case: the estimated alpha is pulled towards zero (equation 5 of the paper)

$$\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}, \qquad \kappa = \frac{1}{1 + \sigma^2/(\tau^2 T)}$$

  - σ: residual volatility
  - τ: dispersion of the true alphas
  - T: number of months
  - short, noisy samples are shrunk more, as in Vasicek (Section 3): σ^2/T plays the role of $\text{se}(\hat{\beta}_i)^2$ and τ^2 that of σ_β^2
- **Replication rate**: among the factors significant in the original studies, the share whose alpha is also significant in the new data
  - in the Bayesian version, a factor is "significant" if its posterior mean divided by its posterior standard deviation is at least 1.96
- Full hierarchy and estimation: Appendix

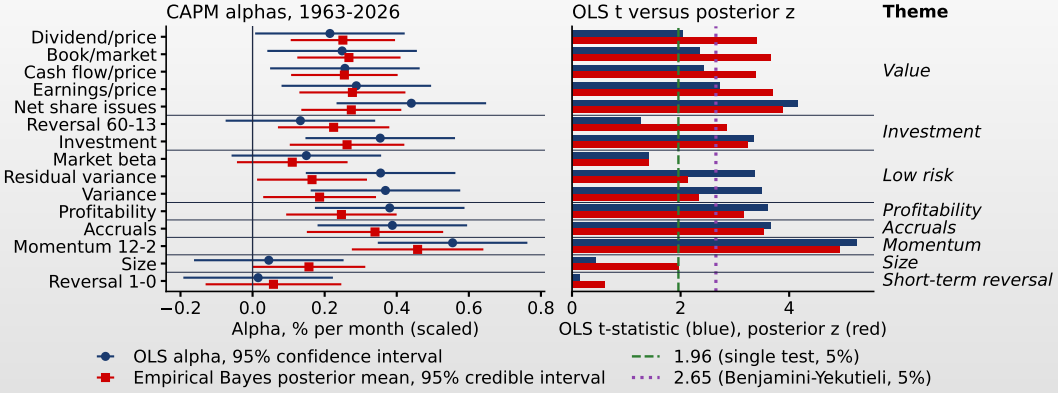
Most Factors Replicate, but Decay after Publication

Replication rates, US (Figure 1)

Hou, Xue & Zhang (2020), raw returns	35.0%
Jensen, Kelly & Pedersen, raw returns	55.6%
Raw returns, 119 originally significant	61.3%
CAPM alpha, OLS	82.4%
CAPM alpha, Benjamini–Yekutieli	75.6%
CAPM alpha, empirical Bayes	82.4%
World, empirical Bayes	82.4%

- From 35% to 82%, step by step, relative to Hou, Xue & Zhang (2020)
  - longer sample, 15 more factors, different construction: 55.6%
  - only the factors significant in the original studies: 61.3%
  - CAPM alphas instead of raw returns, the largest step: 82.4%
- Empirical Bayes also gives 82.4%: pulling towards zero loses some discoveries, information borrowed from the theme adds others, and the effects offset
- The estimated share of false discoveries (Bayesian FDR) is only 0.1%, without raising the t hurdle
- **Decay** after the period of the original study
  - mean alpha: 0.49% a month in sample, 0.26% afterwards (−47%)
  - regressing later alphas on in-sample alphas gives a slope of 0.26 (Figure 8); shrinkage alone would explain a slope of about 0.9
- The strongest factors decay most: a sign of arbitrage or of data mining (Section 12)
- Complete results: Appendix

Do 15 US Factors Replicate? (1/2)



MFM_ch3_jkp_replication

Do 15 US Factors Replicate? (2/2)

- 15 US long–short factors from value-weighted sorts of the Kenneth French Data Library, July 1963 – July 2026 ($T = 757$)
- CAPM alphas are scaled to 10% idiosyncratic volatility
- Factors are grouped by the themes of the paper (authors' cluster labels)
- Replication rates
  - OLS: 73.3% (11 of 15 factors)
  - Benjamini–Yekutieli: 53.3% (8 factors; critical $|t| = 2.65$)
  - Empirical Bayes: 86.7% (13 factors)
- Shrinkage pulls alpha towards the theme mean, not only towards zero
- Example: long-term reversal (investment theme) rises from 0.13% to 0.23% a month; its OLS t is 1.25, but its posterior z is 2.85
- Not replicated: market beta and short-term reversal ($z \leq 1.41$)
- Size only just passes ($z = 1.96$); in the paper too, size is one of the two themes with a rate below 50%
- The rate here uses all 15 signals, without checking whether each was significant in its original study
- Details: Appendix; replication target of Seminar 3, problem C3

MFM_ch3_jkp_replication

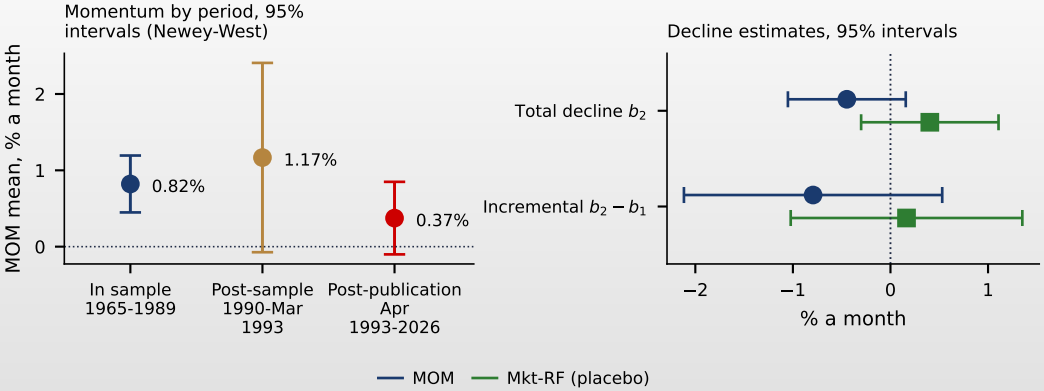
What the Course Replication Cannot Show

- **Limits of the course version**
  - 15 US signals instead of 153 in 93 countries
  - the portfolios of the Kenneth French Data Library, not thirds built without the smallest firms (micro-caps)
  - a single country, so each signal appears once and the signal component cannot be estimated ($\tau_s = 0$)
  - the result depends on the grouping into themes: if all factors were in one theme, the empirical-Bayes rate would be 100%
- **Limits stated by the authors** (Appendix)
  - the CAPM is the only benchmark model
  - all characteristics are built by the same method
  - the prior ignores unpublished factors
- **Open questions** (the paper's data end in 2020)
  - did the alphas and the replication rate hold in 2021–2026?
  - can US factors, used as a prior, improve the estimates of BVB factors, whose histories are short?
- Data and code of the paper: jkpfactors.com
- Course project: Seminar 3, C3

Open Questions and the Discovery Loop

- **Open questions**
  - why do factor premia decay after publication: arbitrage, data mining or a change in risk?
  - does the decay differ across factors and markets?
- Evidence so far
  - McLean & Pontiff (2016): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication
  - the authors attribute the gap to investors trading on the published papers
  - Jensen, Kelly & Pedersen (2023): most factors replicate, but the strongest decay most
  - Chen & Zimmermann (2022): open data and code for the published predictors
- **One pass of the loop**
  - The AI assistant drafts: a literature table with DOIs (Digital Object Identifiers), competing hypotheses and what each predicts, the regression code, a robustness plan
  - The researcher fixes before running: the event dates (Jegadeesh & Titman (1993): sample ends 1989, published March 1993), the list of factors, the multiple-testing correction and the placebo test (a factor where the effect should not appear)
  - Checks: every DOI link resolves
  - The chapter's split-sample t-statistics are reproduced first
  - The data vintage is recorded
- Reusable prompts and the full checklist: Appendix

Mini-Case: Was Momentum Arbitraged Away? (1/2)



MFM_ch3_ai_discovery

Mini-Case: Was Momentum Arbitraged Away? (2/2)

- Design of McLean & Pontiff (2016): $\text{MOM}_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$, monthly, January 1965 – August 2026 ($T = 740$), Newey–West errors
- Post-sample: January 1990 – March 1993
- Post-publication: from April 1993
- PostSample_t, PostPub_t: dummy variables (0/1 indicators) equal to 1 in these periods
- a: the in-sample mean
- b_2: total decline relative to the in-sample mean
- $b_2 - b_1$: incremental effect of publication
- In-sample momentum mean: $a = 0.82\%$ a month ($t = 4.32$)
- Decline after publication: $b_2 = -0.45\%$ ($t = -1.45$, interval $[-1.05\%, 0.16\%]$), a mean 54% lower
- Incremental effect of publication: $b_2 - b_1 = -0.79\%$ ($t = -1.17$)
- Placebo on Mkt-RF: $b_2 = +0.40\%$ ($t = 1.12$)
- The decline is close to the 58% of McLean & Pontiff (2016), but with one factor we cannot reject $H_0 : b_2 = 0$ at 5%
- Arbitrage is one possible interpretation, not a demonstrated cause

MFM_ch3_ai_discovery

Project Design: A Factor Discovery

- **Questions**
  - do the Fama–French factors and momentum decay after publication?
  - does the decay differ across factors?
- **Design**
  - one pooled regression of SMB, HML, RMW, CMA and MOM, with each factor's own dummies for the post-sample and post-publication periods
  - Mkt-RF as placebo
  - errors robust to correlation across factors and to autocorrelation
  - whether the decay is larger where arbitrage is easier can only be tested with stock-level data (size, liquidity, short-sale costs)
- **Replicate first**: the split-sample t-statistics and the mini-case estimates

Project Design: Sample Dates and Deliverables

- **Deliverables**
  - first, an event table: factor, original paper with DOI, end of its sample, publication date, each read from the paper (model row: MOM, Jegadeesh & Titman (1993), sample ends 1989, published March 1993)
  - a pre-registration note (H_0, H_1, dates from the table, list of factors, corrections), saved in Git (a commit) before seeing the results
  - code as a Quantlet: one folder with the script, its data source and a short description, as in the chapter's Quantlets
  - a report with Newey–West errors, placebo periods and the power of the test
  - power: the probability of rejecting $b_2 = 0$ at 5% if the true decline were, for example, the 58% of McLean & Pontiff (2016); by simulation or approximately $\Phi(\delta / \mathrm{se}(b_2) - 1.96)$ (δ: the assumed decline; Φ: the distribution function of the standard Normal distribution)
- **Logs**: prompts in `AI_USE.md`
- Errors caught in `AI_ERRORS.md`

Key Takeaways: An Empirical Workflow

1. **Estimate exposures** with HAC errors, report sample and proxy, shrink extreme betas (Blume, Vasicek)
2. **Test all the alphas at once**: GRS measures how much the Sharpe ratio rises when the test assets are added
   - the CAPM fails on the 25 size $\times$ B/M portfolios (flat SML, GRS $= 4.20$)
3. **Estimate premia** in two passes, with errors that account for estimated betas and for a possibly wrong model
   - a high R^2 on the 25 portfolios proves little
4. **Guard against the zoo**: hurdles corrected for multiple testing, spanning regressions, model comparison by the maximum Sharpe ratio
5. **Replicate**: latent factors and out-of-sample evidence
   - factor ETFs offer cheap factor exposure, not alpha

⊡ Open questions

- why do premia decay after publication?
- how are assets priced when betas vary over time?
- which factors explain expected returns in small markets such as the BVB?

Check Your Understanding

Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not? (XLK example)
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months? (test assets)
- Why does a useless factor look priced in pass 2? (useless factors)
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate? (case study)

Where the ideas continue

- Chapter 2: an alpha is an anomaly or a missing risk factor, the joint hypothesis problem
- Chapter 4: factor covariance models and shrinkage for Markowitz
- Chapters 5–6: time-varying volatility, correlation and betas
- Chapter 8: backtesting factor strategies without overfitting
- Chapter 13: machine learning for the cross section of returns

GRS under the Null Hypothesis

- Under H_0 and the exact assumptions, $W \sim F(N, T-N-1)$, so $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Invert the Sharpe-ratio form of W: the squared Sharpe-ratio gain expected under H_0 is

$$\mathbb{E}\left[\widehat{SR}^2(f, \mathrm{R}) - \widehat{SR}^2(f)\right] \approx \frac{N\left(1 + \widehat{SR}^2(f)\right)}{T-N-3}$$

- 25 portfolios, $T = 757$: $25 \times 1.018/729 = 0.035$ (monthly), against $0.405^2 - 0.134^2 = 0.146$ observed
- Adding N assets always raises the in-sample Sharpe ratio; the test asks whether the rise exceeds what noise would produce
- 9 sector ETFs, $T = 331$: observed gain $0.191^2 - 0.146^2 = 0.015$, below the expected $9 \times 1.021/319 = 0.029$

Back

Tests with Many Assets

- $\hat{\boldsymbol{\Sigma}} = \hat{\mathbf{E}}^\top \hat{\mathbf{E}}/T$ has rank at most $T - 2$ (two parameters per regression)
  - $\hat{\mathbf{E}}$: the $T \times N$ matrix of residuals
  - Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$): $\hat{\boldsymbol{\Sigma}}$ is singular, GRS does not exist (Seminar 3, problem B7)
  - Well before this point the inverse is badly estimated and power falls
- Tests for many assets that avoid $\hat{\boldsymbol{\Sigma}}^{-1}$: standardised sums of squared t-statistics, Pesaran & Yamagata (2024)
- Power enhancement by screening large alphas: Fan, Liao & Yao (2015)

Back

Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects the model here, so γ has no "true" value: we estimate the value that minimises the pricing errors (the pseudo-true value)
- The noise in $\hat{B}$ multiplies e and adds a variance term (Kan, Robotti & Shanken (2013))
- Write both passes as one exactly identified GMM system (GMM in Brief): moments $(R_t - a - Bf_t) \otimes (1, f_t)$ and $X^\top(R_t - X\gamma)$
  - the sandwich variance with the full Jacobian is the robust one
  - Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of Jagannathan & Wang (1998)
- FF5 on the 25 portfolios (RMW: Worked Example slide): the s.e. of CMA rises from 2.2% to 2.7%, that of the intercept from 3.3% to 3.8%
- For HML the change is below 0.05 percentage points
- Advanced course: Generalized Method of Moments

Back

Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): how far the model's SDF is from the nearest SDF that prices all assets correctly

$$\delta^2 = \min_{\mathsf{b}} \; \mathsf{e}(\mathsf{b})^\top \, \mathbb{E}[\mathsf{R}^e \mathsf{R}^{e\top}]^{-1} \, \mathsf{e}(\mathsf{b}), \qquad \mathsf{e}(\mathsf{b}) = \mathbb{E}[m_t(\mathsf{b}) \, \mathsf{R}_t^e]$$

  - Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$
  - For gross returns R_t: $\mathsf{e} = \mathbb{E}[m\mathsf{R}] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
  - The same weighting matrix for every model, so distances are comparable
  - The test statistic has a weighted χ^2 limit
- Selection among very many models: Bryzgalova, Huang & Julliard (2023) average over about 2×10^{15} (two quadrillion) factor models, robust to weak factors
- Kozak, Nagel & Santosh (2020) shrink the SDF coefficients and find that a few principal components of anomaly returns are enough
- Common lesson: compare all models with the same metric and report its uncertainty

Back

Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
  - LASSO: a regression whose penalty sets the coefficients of unimportant variables exactly to zero, so it selects variables
  - Step 1: among hundreds of existing factors, LASSO selects those that explain the average returns of the test assets
  - Step 2: a second LASSO selects those correlated with the new factor, so that omitting them does not distort the result
  - Step 3: the SDF coefficient of the new factor is tested, controlling for all factors selected in steps 1 and 2
- Result: most new factors are redundant
- Only a few new factors explain returns that the old ones do not
- The case study reaches a compatible conclusion: the zoo is highly redundant, and a few themes carry the information

Back

Why the Factor-Number Estimators Disagree

- Onatski (2010): a factor is real if its eigenvalue stands clearly outside the range where the noise eigenvalues fall
- Kaiser ($\ell_k > 1$) on a correlation matrix: with N assets and pure noise the eigenvalues spread around 1 with a width that depends on N/T
- Bai–Ng with $N = 11$ and $T = 2073$: the penalty is $\frac{N+T}{NT} \ln 11 = 0.219$ per factor
- On the sectors, the criterion $IC_{p2}(k)$ for $k = 0, \ldots, 5$ is 0, -0.86, -0.97, -1.12, -1.14, -1.17
- It falls all the way to the maximum allowed, $k_{\max} = 5$: the penalty is too small to stop weak factors from being added
- Report the range of estimates and the sensitivity to $k_{\max}$, not a single number (Seminar 3, problem B3)

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Risk-Premium PCA and Three-Pass Premia

- **RP-PCA** (Risk-Premium PCA, Lettau & Pelger (2020)): principal components of $\frac{1}{T}\mathbf{X}^\top\mathbf{X} + \gamma\bar{\mathbf{x}}\bar{\mathbf{x}}^\top$
  - $\mathbf{X}$: the $T \times N$ matrix of excess returns, $\bar{\mathbf{x}}$: their means, γ: the weight on average returns
  - $\gamma = -1$ gives covariance PCA
  - $\gamma > -1$ also favours factors that explain average returns
  - In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- **Three-pass estimator** (Giglio & Xiu (2021)): the premium of an observed factor g_t when the model omits factors
  - (1) PCA of the test-asset returns
  - (2) cross-sectional regression of the means on the latent betas
  - (3) regression of g_t on the latent factors; the premium of g_t follows from the latent premia and the coefficients of this regression
- Both keep the statistical basis of PCA but aim to explain average returns, not only variance
- IPCA also adds the characteristics of the assets

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IPCA and Crypto-Asset Factors

- IPCA (Kelly, Pruitt & Su (2019)): characteristics predict returns because they measure risk exposure
- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of cryptocurrencies (Liu, Tsyvinski & Wu (2022))
- Dimension reduction on the return features of crypto and traditional assets: Pele et al. (2023), Jheng et al. (2026)
- Advanced course: Multivariate Statistical Analysis

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Building the 153 Factors

- **Data** (Section II): 153 factors in 93 countries
  - 138 from the list of Hou, Xue & Zhang (2020), plus 15 from the wider literature
  - US returns from 1926, most developed markets from 1986
  - all in US dollars, in excess of the US Treasury bill
- **Construction**, each country and month
  - thirds by the characteristic, with breakpoints computed without the smallest firms (micro-caps)
  - "capped" value weights: market equity winsorised at the NYSE 80th percentile
  - one-month holding period
  - at least 5 stocks per leg and 60 monthly observations
- **Themes**: 13 clusters of US factors (Section II.B)
  - hierarchical clustering, with distance 1− correlation of CAPM residuals and Ward's method (it merges the groups that least increase the variance within groups)

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The Bayesian Hierarchy of Alphas

- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
  - alphas $\alpha_i = c_j + s_n + w_i$ (theme j, signal n, factor i; eq. 21), prior mean 0
  - prior variance $\mathbf{\Omega} = \mathrm{M}\mathrm{M}^\top \tau_c^2 + \mathrm{Z}\mathrm{Z}^\top \tau_s^2 + \mathrm{I}\tau_w^2$ (eq. 23)
  - posterior mean $(\mathbf{\Omega}^{-1} + T\mathbf{\Sigma}^{-1})^{-1} T\mathbf{\Sigma}^{-1}\hat{\boldsymbol{\alpha}}$ (Proposition 4)
  - M, Z: 0/1 matrices linking each factor to its theme and signal
  - $\tau_c^2, \tau_s^2, \tau_w^2$: variances of the three alpha components
  - $\mathbf{\Sigma}$: residual covariance
  - empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\boldsymbol{\alpha}} \sim N(0, \mathbf{\Omega} + \hat{\mathbf{\Sigma}}/T)$ (Section I.C.2)
- **Replication rate**
  - share of the factors significant in the original studies whose alpha is significant in the new data

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Complete Results

- Raw returns, restricted to the 119 factors significant in the original studies: 61.3% (Figure 1)
- With OLS, 98 of the 119 US factors replicate
- The Benjamini–Yekutieli correction requires $|t| > 2.7$ (Figure 4)
- Bayesian FDR: 0.1%, with 95% interval $[0, 1\%]$
- Expected share of true factors: 94% (Section III.C)
- Replication rate for mega and large stocks: 77.3% and 79.8% (Figure 5)
- 11 of 13 themes have a rate of at least 50%; the exceptions are low leverage and size
- The high rate among large firms shows that the result does not come only from the smallest firms
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

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The Course Replication in Detail

- Benjamini–Yekutieli gives the lowest rate, as in the US results of the paper (Figure 1)
- Net share issues are pulled towards the mean of their theme: from 0.44% to 0.27% a month
- The posterior standard deviation is 0.07%–0.10%, below the OLS s.e. of 0.11%: each factor borrows information from the other factors of its theme
- **Prior dispersion**: $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month
  - against 0.35% and 0.21% in Table I of the paper
  - with 15 signals little factor-specific dispersion is left
- **Themes**: the 15 signals fall into 8 of the paper's 13 themes
  - 5 in value, 3 in low risk, 2 in investment

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Limits Stated by the Authors and Extensions

- **Limits stated by the authors**
  - The CAPM is the only benchmark model: low-leverage factors have no significant CAPM alpha, but contribute when combined with the other themes (Section III.A)
  - All characteristics are built by the same rule, sometimes differently from the original paper (Section II.C)
  - Factors tried but never published would shift the prior: in the scenarios of Harvey, Liu & Zhu (2016) the US rate falls to 81.5% and 79.8% (Figure 10)
- **Complete the course replication** (Seminar 3, C3)
  - justify each sign from the original paper
  - compare the dendrogram with the paper's themes
  - add the Bayesian FDR and the post-sample slope (Figure 8)
- Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis

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The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.`
  - Resolve every DOI on Crossref
  - Check each headline number against the abstract
- **Hypothesis**
  - `Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.`
- **Data and code**
  - `Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.`
  - Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

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The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).`
- **Critique**
  - `Referee this: one factor, one publication date, t = -1.45. What is the power to detect a 58% decay? Why does pooling many anomalies change it?`
- **Report**
  - In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
  - Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
  - Prompts that shaped a decision go into `AI_USE.md`
- The assistant enumerates the specifications
- The researcher fixes in advance which ones count

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What the Researcher Must Check

- **Wrong event date**: sample end, working paper and journal issue are different dates
  - test: read them from the paper itself
  - Jegadeesh & Titman (1993): sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay**: fix the list of factors in advance
- Apply Holm or BH across all factors
- **Serially correlated, crash-prone returns**: Newey–West errors
- Re-estimate without 2009
- **Data vintages**: the factor library is revised retroactively
  - test: record the download date
  - reproduce the chapter's split-sample t-statistics
- **Misquoted literature**: a common slip gives 58% as the out-of-sample decline
- In fact 58% is the post-publication decline; out of sample the decline is 26%
- **Low power**: with a single factor, H_0 is rarely rejected
- Report the confidence interval of the decay, not only its sign

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Answers to the Quick Checks (1/2)

Beta, error and shrinkage

- Classical: $0.90 \pm 1.96 \times 0.05 = [0.80, 1.00]$, just below 1; Newey–West: $0.90 \pm 0.18 = [0.72, 1.08]$, contains 1: no evidence that the stock is less risky
- Report Newey–West: residuals are heteroskedastic and autocorrelated; HAC changes the estimated variance, not the OLS estimate
- $\sigma^2_\beta = 0.10 - 0.02 = 0.08$; s.e. 0.2: $w = 0.08/0.12 = 0.67$, $\beta^V = 0.67 \times 1.6 + 0.33 \times 1.0 = 1.40$; s.e. 0.1: $w = 0.08/0.09 = 0.89$, $\beta^V = 1.53$
- The noisier estimate is shrunk more; Blume gives $0.33 + 0.67 \times 1.6 = 1.40$ whatever the precision

Reading a GRS statistic

- $W = \frac{60}{4} \times (0.25^2 - 0.15^2)/(1 + 0.15^2) = 15 \times 0.040/1.0225 = 0.59 < 2.53$: not rejected
- Expected gain $4 \times 1.0225/58 = 0.071$, above the observed 0.040: the gap is within noise
- $T = 650$: $W = \frac{645}{4} \times 0.0391 = 6.31 > 2.39$: rejected; the gap is the same, the evidence grows with T

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Answers to the Quick Checks (2/2)

Premia, Shanken and a negative R^2

- $\text{se}_{FM} = 3\%/\sqrt{400} = 0.15\%$, $t = 3.33$
- $c = 0.5^2/1^2 = 0.25$; $\text{Var}_{\text{Sh}} = 1.25 \times (0.0225 - 0.0025) + 0.0025 = 0.0275$, s.e. 0.166%, $t = 3.02$: only just above 3
- Traded factor: $c = 0.25/4.5^2 = 0.012$, so the correction is negligible
- Mean 4%: $\text{SST} = 4 + 0 + 4 = 8$, $\text{SSE} = 16 + 0 + 16 = 32$, $R^2 = 1 - 32/8 = -3$; with a free intercept, zero slopes already give SSE = SST, and OLS can only lower SSE

Counting discoveries

- Naive 4; Bonferroni (0.01) 1; Holm: $0.004 \leq 0.01$, $0.012 \leq 0.0125$, $0.020 > 0.0167$ stop: 2; BH thresholds $0.01j$: $0.030 \leq 0.040$ is the last success: 4
- BY: the sum is 2.283, thresholds $0.0044j$: only 0.004 passes: 1
- $1 - 0.95^5 = 0.23$; a search over candidates: BH (FDR); "no factor is spurious": Holm or Bonferroni (FWER)

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Further Study: Courses and Code

Advanced courses on quantinar.com

- Generalized Method of Moments
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- Factor data: Kenneth French Data Library

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Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **APT**: Arbitrage Pricing Theory
- **BAB**: Betting Against Beta
- **BET**: Bucharest Exchange Trading (index)
- **BET-TR**: BET Total Return (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **BY**: Benjamini–Yekutieli (procedure)
- **CAGR**: Compound Annual Growth Rate
- **CAPM**: Capital Asset Pricing Model
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CML**: Capital Market Line
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model

Acronyms used in this material (2/3)

- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HML**: High Minus Low (value factor)
- **IPCA**: Instrumented Principal Component Analysis
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MOM**: Momentum (factor)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **PC1**: first Principal Component
- **PC2**: second Principal Component
- **PC3**: third Principal Component
- **PCA**: Principal Component Analysis

Acronyms used in this material (3/3)

- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **RF**: Risk-Free rate
- **RMSE**: Root Mean Squared Error
- **RMW**: Robust Minus Weak (profitability factor)
- **ROE**: Return On Equity
- **RP-PCA**: Risk-Premium Principal Component Analysis
- **SDF**: Stochastic Discount Factor
- **SMB**: Small Minus Big (size factor)
- **SML**: Security Market Line
- **SSE**: Sum of Squared Errors
- **SST**: Total Sum of Squares
- **UMD**: Up Minus Down (momentum factor)
- **US**: United States